

1.2 Converting Between Repeating Decimals and Fractions

Lesson Introduction

 Benchmark	MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.		 Learning Target	I can rewrite rational numbers in different but equivalent forms, including repeating decimals.
 Benchmark Coherence	Building On		Working Toward	
	6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.		MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.	
 Essential Questions	How can you convert repeating decimals to fractions and percentages?			
 Lesson Narrative	In this lesson, students start by activating prior knowledge from Lesson 1 of converting a fraction to a repeating decimal using technology. Students then learn how to rewrite repeating decimals as fractions using equations and base-10 reasoning. Students then apply this knowledge to converting repeating decimals into equivalent percentages and fractions.			
 Component Summary	1.2.1 Warm-Up (~5 min)	Activate prior knowledge of repeating decimals (<i>SE p. 13</i>)	1.2.4 Collaboration (10 min)	Repeating decimals as fractions and percentages (<i>SE p. 15</i>)
	1.2.2 Guided Instruction (10 min)	Converting repeating decimals to fractions using an equation (<i>SE p. 13</i>)	1.2.5 Your Turn! (10 min)	Practicing rewriting repeating decimals as fractions and percentages (<i>SE p. 16</i>)
	1.2.3 Guided Practice (5–10 min)	Practice rewriting repeating decimals as fractions (<i>SE p. 14</i>)	1.2.6 Wrap-Up (~5 min)	Converting between decimal, percentage and fraction (<i>SE p. 17</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Repeating - Terminating - Percentage - Sample space		(Optional) Calculators	
				Additional Preparation
				N/A

 Teacher Tips	Common Misconceptions • Students may mix up the order when dividing the numerator and denominator of the fractions. • Students may struggle with the equation used to change repeating decimals into fractions and recognizing which equations to use to eliminate the repeating pattern. • Students may forget to add the bar above the digits in a repeating decimal.	Things to Consider • Consider allowing students to use calculators to verify their work after rewriting fractions as repeating decimals.
	Literacy and Language Routines Algebra Talk In 1.2.2 Guided Instruction and 1.2.3 Guided Practice, prompt students to discuss the similarities and differences between the equations created, particularly the repeating portions of the decimals. “Algebra talk” builds algebraic thinking by encouraging students to think about the numbers in an expression and what they know about structure, patterns, and properties of operations to mentally solve a problem. This will be helpful when approaching similar problems, as there is not always a set algorithm to use for this process.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1 Mathematical Fluency Students are provided with the opportunity to complete tasks with mathematical fluency. In this lesson, students learn to rewrite repeating decimals into fractions and percentages with mathematical fluency.
	 Differentiate Instruction	If students are struggling with generating and solving the equations used to rewrite repeating decimals as fractions, consider providing blanks to fill in in order to prompt students’ thinking. Consider also making explicit connections between the patterns students observe after each component.
Lesson Notes:		

1.2.1 Warm-Up: Error Analysis

Component Overview: Students analyze a student error after they misinterpreted a value when converting a fraction to a repeating decimal using a calculator.



1.2.1 Warm-Up: Error Analysis

1. Isaiah used a scientific calculator to find the decimal equivalent of $\frac{16}{6}$. The calculator screen displayed "2.666666667". Isaiah wrote the following equation on his paper:

$$\frac{16}{6} = 2.666666667$$

- a. Explain why Isaiah's equation is incorrect.

Answers will vary. Isaiah's equation is incorrect because the decimal equivalent of the fraction is the repeating decimal, $2.\bar{6}$, and Isaiah wrote it as a terminating decimal.

- b. What advice would you give Isaiah when using a calculator to find decimal equivalents in the future?

Answers will vary. I would advise Isaiah to find the decimal equivalents by using long division and compare the answers with the calculator answers to see if the decimals are repeating or terminating.



The Warm-Up has students thinking about a common mistake that they might make. Prompt students to use long division and see if the result is a repeating pattern.

1.2.2 Guided Instruction: Rewriting Repeating Decimals as Fractions

Component Overview: Students learn how to convert a repeating decimal to a fraction using equations and base-10 reasoning.



1.2.2 Guided Instruction: Rewriting Repeating Decimals as Fractions

When given a repeating decimal, equations can be used to find its equivalent value represented as a fraction.

1. Consider the decimal $0.0\overline{6}$. Some fraction is equivalent to $0.0\overline{6}$, which can be represented by the equation, $x = 0.0\overline{6}$.

- a. If $x = 0.0\overline{6}$, complete the step shown to determine what value is equivalent to $10x$.

$$10 \cdot x = 0.0\overline{6} \cdot 10$$

$$10x = \boxed{0.\overline{6}}$$

- b. Determine what value is equivalent to $100x$.

$$100 \cdot x = 0.0\overline{6} \cdot 100$$

$$100x = \boxed{6.\overline{6}}$$

- c. These relationships can be used to find the value of x represented as a fraction. Complete the steps shown.

$$100x - 10x = \boxed{6.\overline{6}} - \boxed{0.\overline{6}}$$

$$90x = 6$$

$$x = \frac{\boxed{6}}{\boxed{90}}$$

- d. Rewrite the solution as a fraction in simplest form.
- $$\frac{1}{15}$$



Students often struggle with this abstraction. Consider modeling the first question using the Think-Aloud strategy, allowing opportunities for students to ask questions. Then question 2 can be guided, where students are released to do one part at a time, followed by a debrief to ensure understanding.



Algebra Talk

Consider highlighting that when subtracting $6.\overline{6} - 0.\overline{6}$, it resulted in the repeating portion of the decimal being eliminated. This is part of the reasoning students will need to use when determining which equations to use. Consider showing this on the calculator as well.

1.2.2 Guided Instruction: Rewriting Repeating Decimals as Fractions (continued)

2. Consider the repeating decimal $0.\overline{36}$.

a. If $x = 0.\overline{36}$, complete each equation.

$$x = 0.\overline{36}$$

$$10x = \underline{3.\overline{63}}$$

$$100x = \underline{36.\overline{36}}$$

b. Which two values, when subtracted, would result in the repeating portion of the decimal being eliminated?

$36.\overline{36}$, $0.\overline{36}$

c. Use two of the relationships represented in the equations in Part A to complete the steps to find the fraction equivalent of $0.\overline{36}$.

$$100x - x = 36.\overline{36} - 0.\overline{36}$$

$$99x = \underline{36}$$

$$x = \frac{\boxed{36}}{\boxed{99}}$$

d. Rewrite the solution as a fraction in simplest form, if possible.

$$\frac{4}{11}$$



When turning a repeating decimal into a fraction, connect some of the patterns that students have observed:

- $0.aaaaaaa... \text{ to } \frac{a}{9}$.
- $0.abababab... \text{ to } \frac{ab}{99}$.

1.2.3 Guided Practice: Rewriting Repeating Decimals as Fractions

Component Overview: Students practice rewriting repeating decimals as fractions using an equation and base-10 reasoning.



1.2.3 Guided Practice: Rewriting Repeating Decimals as Fractions

1. Consider the repeating decimal $0.3\bar{8}$.

- a. If $x = 0.3\bar{8}$, complete each equation.

$$x = 0.3\bar{8}$$

$$10x = \underline{3.\bar{8}}$$

$$100x = \underline{38.\bar{8}}$$

- b. Which two values, when subtracted, would result in the repeating portion of the decimal being eliminated?

$38.\bar{8}$, $3.\bar{8}$

- c. Use two of the relationships represented in the equations in Part A to complete the steps to find the fraction equivalent of $0.3\bar{8}$.

$$100x - 10x = 38.\bar{8} - 3.\bar{8}$$

$$90x = \underline{35}$$

$$x = \frac{\boxed{35}}{\boxed{90}}$$

- d. Rewrite the solution as a fraction in simplest form, if possible.

$$\frac{7}{18}$$

- e. Write the fraction or mixed number that is equivalent to $14.3\bar{8}$.

$$14\frac{7}{18}$$



Algebra Talk

Prompt students to discuss the similarities and differences between the equations, particularly the repeating portions of the decimals. “Algebra Talk” builds algebraic thinking by encouraging students to think about the numbers in an expression and what they know about structure, patterns, and properties of operations to mentally solve a problem. This will be helpful when approaching similar problems, as there is not always a set algorithm to use for this process.

1.2.4 Collaboration: Fractions, Decimals, and Percentages

Component Overview: Students work together to convert between fractions, decimals, and percentages with repeating decimals.



1.2.4 Collaboration: Fractions, Decimals, and Percentages

Work with your partner to complete the following.

1. In a standard deck of Uno cards, there are 24 action cards that include "Skip," "Draw Two," and "Reverse." There are 108 cards in an Uno deck.

- a. What fraction of the Uno cards are action cards?

$$\frac{6}{27}$$

- b. Express your answer to part a as a decimal and a percentage.

Decimal	Percent
0. $\bar{2}$	22. $\bar{2}$ %

- c. Discuss with your partner why 22% would be an incorrect equivalent value and summarize your discussion in the space below.

Answers will vary. It would be incorrect because the correct equivalent is a repeating decimal and 22% is not a repeating decimal. This is a rounded percentage rather than the exact percentage.

2. Aliyah played a game using one standard die. There is a 33. $\bar{3}$ % chance of winning the game on any roll.

- a. Express the chances of winning the game as a fraction and a decimal.

Decimal	Fraction
0. $\bar{3}$	$\frac{1}{3}$



Consider a review of converting from a decimal/fraction to a percentage, if needed.

- How do you convert from a decimal to a percentage?
- How do you convert from a fraction to a percentage?

1.2.4 Collaboration: Fractions, Decimals, and Percentages (continued)

- b. Which representation – decimal, fraction, or percentage – is the easiest way to describe the chance of a winning roll?

Answers will vary. The fraction is the easiest way because it communicates 1 out of 3 chances to win the clearest.

- c. What are the possible results in the sample space when rolling a standard six-sided die?

1, 2, 3, 4, 5, 6

- d. Discuss with your partner what you think the rules of Aliyah's game might be. Then, complete the statement with one possibility.

To win the game, Aliyah must roll

Answers will vary. The rules might be for a player to win, they can roll a factor of 2 (or 3 or 5), a number divisible by 3, a composite number, a multiple of 3, a number less than 3, a number greater than 4, a 6 or a 1 (or any other combination of two numbers).



Think-Pair-Share

For question 2b, have students engage in a Think-Pair-Share. Students should describe which representation they think is the easiest way to describe the chance of winning. Then they can discuss with their partner before sharing with the class.

1.2.5 Your Turn!

Component Overview: Students apply what they have learned so far to practice converting between equivalent forms of repeating decimals, fractions, and percentages.



1.2.5 Your Turn!

1. In a package of Skittles, Carmela found that 21% of them were red and $0.1\bar{6}$ were green.

- a. What fraction of the Skittles was red?

$$\frac{21}{100}$$

- b. What fraction of the Skittles was green?

$$\frac{1}{6}$$

- c. She found 17 of the 60 skittles are purple. What percent of the Skittles are purple?

$$28.\bar{3}\%$$



For students who are struggling with questions 1b, remind students to review 1.2.2 Guided Instruction. Students can check their work by doing long division if needed.

1.2.6 Wrap-Up: Ticket Out the Door

Component Overview: Students practice converting between equivalent forms of decimals, fractions, and percentages.



1.2.6 Wrap-Up: Ticket Out the Door

1. Complete the table.

Decimal	Percent	Fraction	Mixed Number, if Possible
$0.\overline{63}$	$63.\overline{63}\%$	$\frac{7}{11}$	
$0.\overline{12}$	$12.\overline{12}\%$	$\frac{4}{33}$	
$5.\overline{3}$	$533.\overline{3}\%$	$\frac{16}{3}$	$5\frac{1}{3}$
$9.22\overline{72}$	$922.\overline{72}\%$	$\frac{203}{22}$	$9\frac{5}{22}$



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.